

BPHO ROUND 0 · QUESTION PAPER

Drill bank section 1

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking*Original questions written for this course, in the Round 0 format. Not BPhO material.***Before you start**

- Answer every question. There is no negative marking, so a blank is a wasted mark.
- No calculator. Every answer here is reachable by hand, and the numbers are chosen to divide cleanly.
- Work on paper, then mark yourself against the markscheme PDF.
- The 2025 UK qualifying line was 11 out of 25, so about 44% is the target to beat.

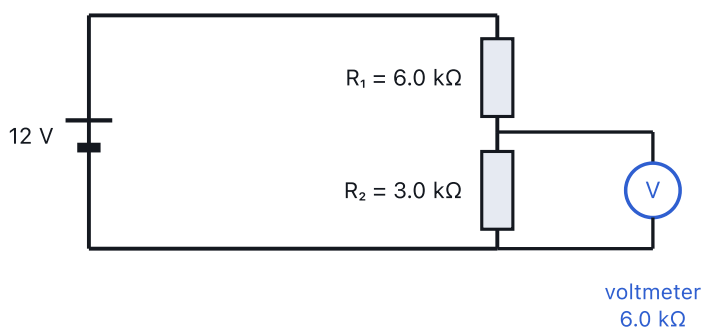
Q1 · Atwood machine: the tension in a string that is not in equilibrium **C**

Two blocks of mass m and $2m$ are joined by a light inextensible string that passes over a smooth fixed pulley, and hang freely. What is the tension in the string while the blocks are moving?

- A** mg **B** $4mg/3$
C $3mg/2$ **D** $2mg/3$
E $8mg/3$

Q2 · A potential divider measured with a voltmeter that is not ideal **H**

A 12 V supply of negligible internal resistance is connected in series with a $6.0 \text{ k}\Omega$ resistor and a $3.0 \text{ k}\Omega$ resistor. A voltmeter of resistance $6.0 \text{ k}\Omega$ is connected across the $3.0 \text{ k}\Omega$ resistor.



What does the voltmeter read?

- A** 1.5 V **B** 2.0 V
C 3.0 V **D** 4.0 V
E 6.0 V

Q3 · Dimensional analysis: the terminal speed of a raindrop against its radius **A**

A spherical raindrop falls through still air and reaches a terminal speed v . The drag force on the drop depends only on the drop's radius r , the density of the air ρ and v itself. The drop is much denser than the air, and the air is much less dense than water (density ρ_w).

How does v depend on r ?

A $v \propto r$

B $v \propto \sqrt{r}$

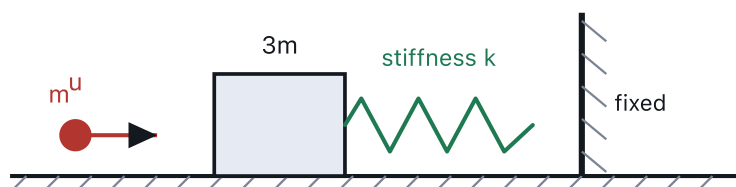
C $v \propto r^2$

D $v \propto 1/\sqrt{r}$

E v does not depend on r

Q4 · An inelastic collision followed by a spring: two conservation laws in a row **C**

A bullet of mass m travelling horizontally at speed u strikes and embeds itself in a block of mass $3m$ resting on a frictionless horizontal surface. The block is attached to a fixed spring of stiffness k .



What is the maximum compression of the spring?

A $u\sqrt{(m/k)}$

B $(u/4)\sqrt{(m/k)}$

C $(u/4)\sqrt{(2m/k)}$

D $2u\sqrt{(m/k)}$

E $(u/2)\sqrt{(m/k)}$

Q5 · Maximum power in a variable load driven by a cell of internal resistance **H**

A cell of emf E and internal resistance r is connected to a variable resistor of resistance R . What is the largest power that can be dissipated in the variable resistor?

A E^2/r

B $E^2/(2r)$

C $E^2/(8r)$

D $E^2/(4r)$

E $E^2 r$

Q6 · Estimation: the mass of the air standing on a square metre **A**

The atmospheric pressure at sea level is about 1.0×10^5 Pa, and **g** is about 10 N kg^{-1} .

Which of these is the best estimate of the mass of the air standing on each square metre of the Earth's surface?

A 10^3 kg

B 10^4 kg

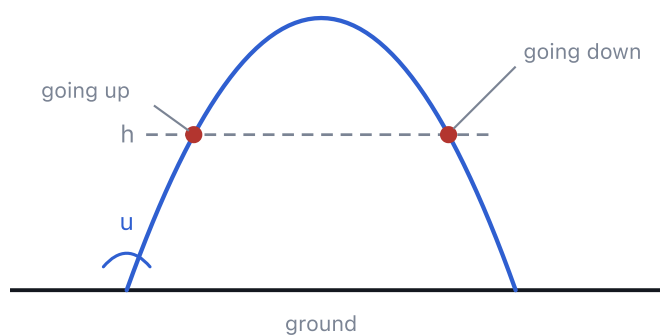
C 10^5 kg

D 10^6 kg

E 10^7 kg

Q7 · A projectile passing the same height on the way up and on the way down **B**

A ball is thrown vertically upward from ground level with speed **u**. It passes a point at height **h** twice: once on the way up and once on the way down.



What is the time interval between the two passages?

A $\sqrt{(u^2 - 2gh)}/g$

B u/g

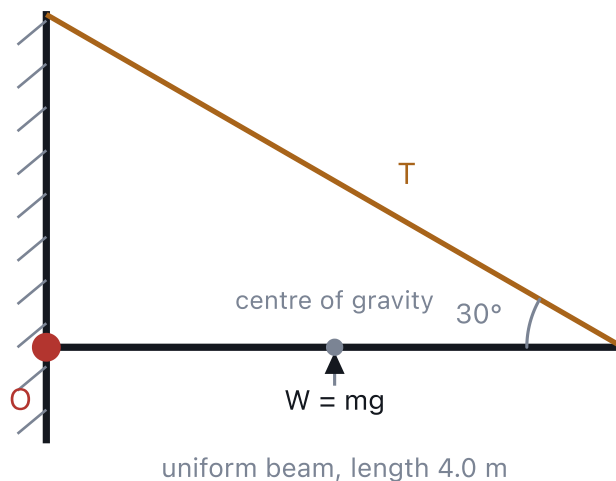
C $2\sqrt{(2gh)}/g$

D $2u/g$

E $2\sqrt{(u^2 - 2gh)}/g$

Q8 · A horizontal beam held by an angled stay: moments about the hinge **C**

A uniform horizontal beam of mass 20 kg and length 4.0 m is freely hinged to a vertical wall at one end. A light stay wire is attached to the other end and runs to the wall, making an angle of 30° with the beam.



What is the tension in the stay?

- | | |
|----------------|----------------|
| A 50 N | B 100 N |
| C 200 N | D 115 N |
| E 400 N | |
-

Q9 · The speed at which a car leaves the road on a hump-backed bridge **D**

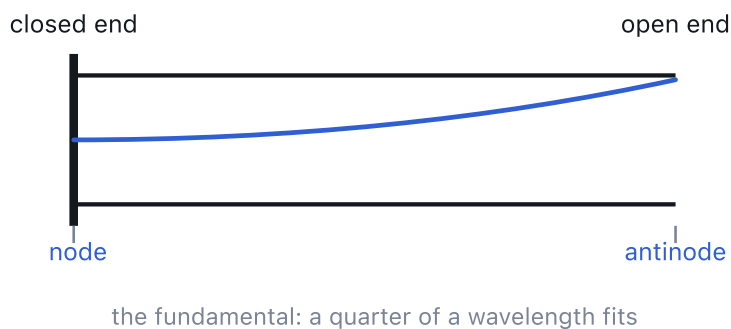
A car crosses a hump-backed bridge whose top may be taken as an arc of radius 20 m. Take $g = 10 \text{ N kg}^{-1}$.

Above what speed does the car leave the road at the top of the bridge?

- | | |
|------------------|------------------|
| A 14 m/s | B 10 m/s |
| C 20 m/s | D 200 m/s |
| E 400 m/s | |
-

Q10 · Three successive resonances of a pipe closed at one end F

A pipe closed at one end is found to resonate at 300 Hz, 500 Hz and 700 Hz, with no resonance in between.

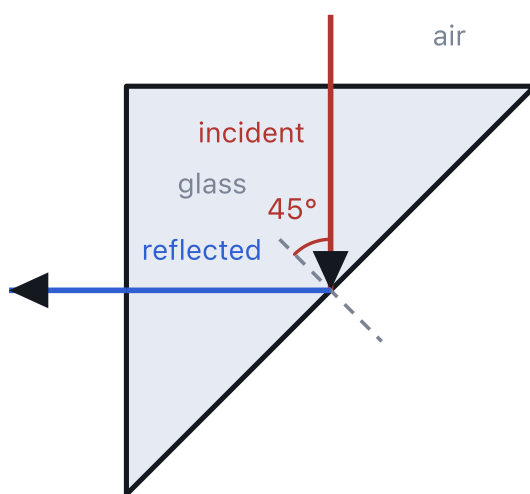


What is the fundamental frequency of the pipe?

- | | |
|-----------------|-----------------|
| A 50 Hz | B 150 Hz |
| C 200 Hz | D 100 Hz |
| E 300 Hz | |

Q11 · The minimum refractive index for total internal reflection inside a prism G

A ray of light enters a glass prism normally through one of the short faces. The prism is a right-angled isosceles triangle, so the ray meets the long face at 45° .



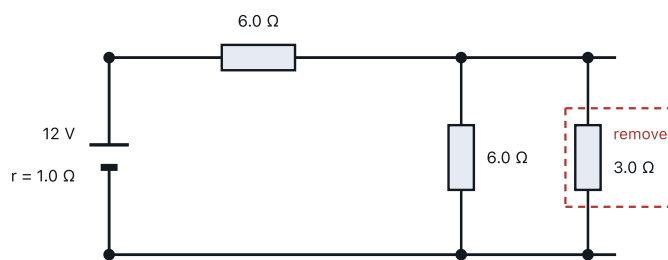
the long face is the hypotenuse

What is the smallest refractive index the glass can have if the ray is to be totally internally reflected at the long face?

- | | |
|---------------------|----------------|
| A 1.2 | B $4/3$ |
| C $\sqrt{2}$ | D 1.5 |
| E $\sqrt{3}$ | |

Q12 · Removing one branch of the load: the ratio of the two terminal potential differences H

The figure shows a cell of emf 12 V and internal resistance $1.0\ \Omega$ driving a load made of a $6.0\ \Omega$ resistor in series with a parallel pair of $6.0\ \Omega$ and $3.0\ \Omega$.



the $3.0\ \Omega$ branch is taken away; the $6.0\ \Omega$ of the pair stays

The $3.0\ \Omega$ resistor is now removed, so the cell drives the two $6.0\ \Omega$ resistors in series. Nothing else is changed.

What is the ratio (terminal p.d. after the removal) / (terminal p.d. before the removal) ?

- | | |
|----------------|---------------|
| A 27/26 | B 9/13 |
| C 3/2 | D 1 |
| E 26/27 | |

Q13 · Thermal equilibrium between a hot metal block and cooler water J

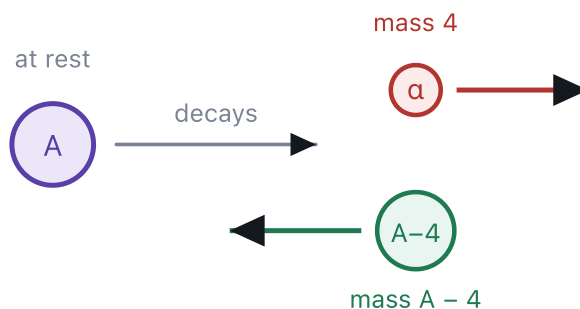
A metal block of mass $0.20\ \text{kg}$ and specific heat capacity $500\ \text{J kg}^{-1}\ \text{K}^{-1}$ is at $200\ ^\circ\text{C}$. It is dropped into $0.10\ \text{kg}$ of water, of specific heat capacity $4000\ \text{J kg}^{-1}\ \text{K}^{-1}$, at $20\ ^\circ\text{C}$. The container is an ideal insulator.

What is the final temperature of the block and water?

- | | |
|--------------------------------|--------------------------------|
| A $20\ ^\circ\text{C}$ | B $40\ ^\circ\text{C}$ |
| C $110\ ^\circ\text{C}$ | D $140\ ^\circ\text{C}$ |
| E $56\ ^\circ\text{C}$ | |

Q14 · Alpha decay: how the released energy is divided between the two products **K**

A nucleus of mass number **A** is at rest and decays by emitting an alpha particle. The total kinetic energy given to the two products is **E**.



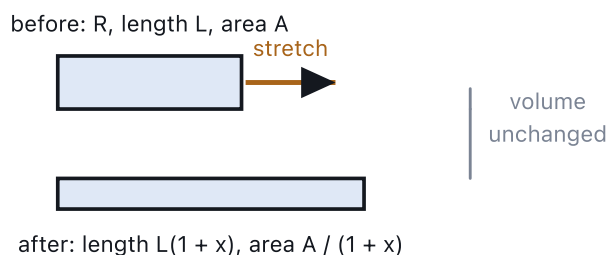
the two products separate with equal and opposite momenta

What is the kinetic energy of the alpha particle?

- A** $E(A - 4)/A$ **B** $4E/A$
C $4E/(A - 4)$ **D** $E/2$
E E

Q15 · Small change: the resistance of a wire stretched at constant volume **A**

A uniform wire of resistance **R** is stretched so that its length increases by a small fraction **x**. The volume of the wire does not change.



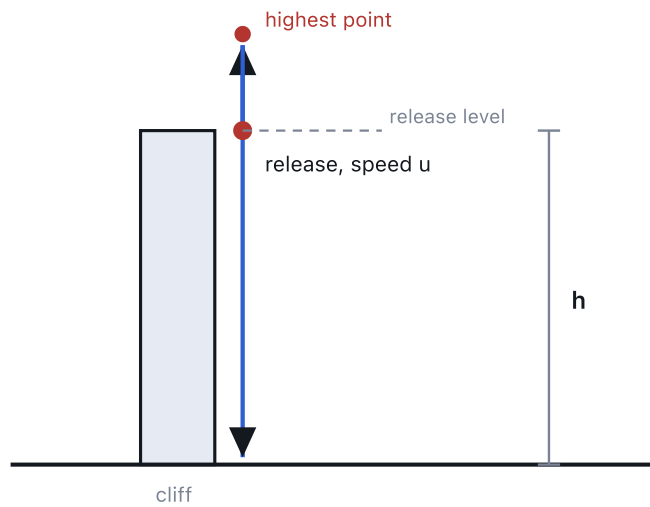
the wire gets longer AND thinner, and both raise the resistance

What is the fractional change in the resistance?

- A** x **B** $2x$
C $x/2$ **D** $4x$
E 0

Q16 · A stone thrown upward from a cliff top: the time to reach the foot as an expression **B**

A stone is thrown vertically upward with speed **u** from the top of a vertical cliff of height **h**. It rises, falls back past the point of release, and lands at the foot of the cliff. The acceleration due to gravity is **g**.



Which expression gives the time from release to landing?

A $(u + \sqrt{u^2 + 2gh}) / g$

B $\sqrt{u^2 + 2gh} / g$

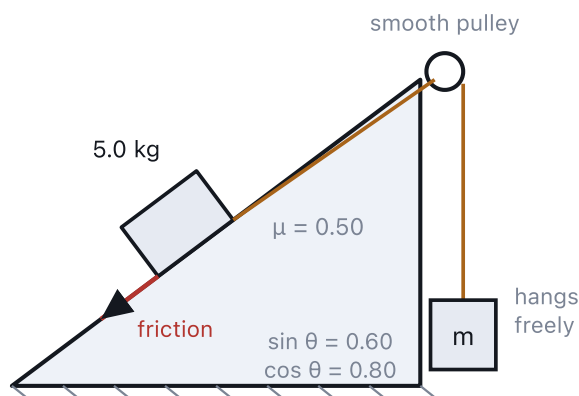
C $u/g + \sqrt{2h/g}$

D $2u/g + \sqrt{2h/g}$

E $\sqrt{2h/g}$

Q17 · Limiting friction: the hanging mass that just starts to move a block **C**

A block of mass 5.0 kg rests on a rough plane inclined at an angle whose sine is 0.60 and cosine is 0.80. The coefficient of friction between block and plane is 0.50. A light string runs from the block, up the slope and over a smooth pulley at the top, and hangs vertically with a mass m on the end.



Take $g = 10 \text{ N kg}^{-1}$. What is the smallest m for which the block just starts to move up the slope?

- | | |
|-----------------|-----------------|
| A 2.0 kg | B 3.0 kg |
| C 4.0 kg | D 5.0 kg |
| E 7.0 kg | |
-

Q18 · Spinning a space station so the rim feels like the Earth's surface **D**

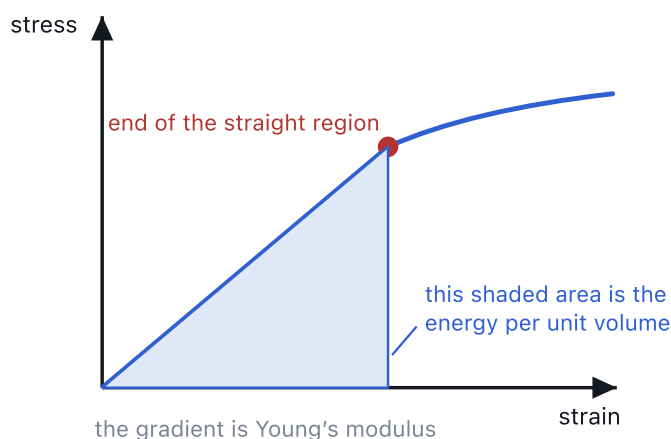
A cylindrical space station of radius 50 m spins about its axis. Take $g = 10 \text{ N kg}^{-1}$.

What period of rotation would make the apparent gravity at the rim equal to g ?

- | | |
|----------------|----------------|
| A 1.4 s | B 2.8 s |
| C 14 s | D 20 s |
| E 31 s | |
-

Q19 · Strain energy per unit volume as the area under a stress-strain graph **E**

The graph shows the stress-strain curve for a metal. The straight-line region ends at a stress of 3.0×10^8 Pa and a strain of 1.5×10^{-3} .

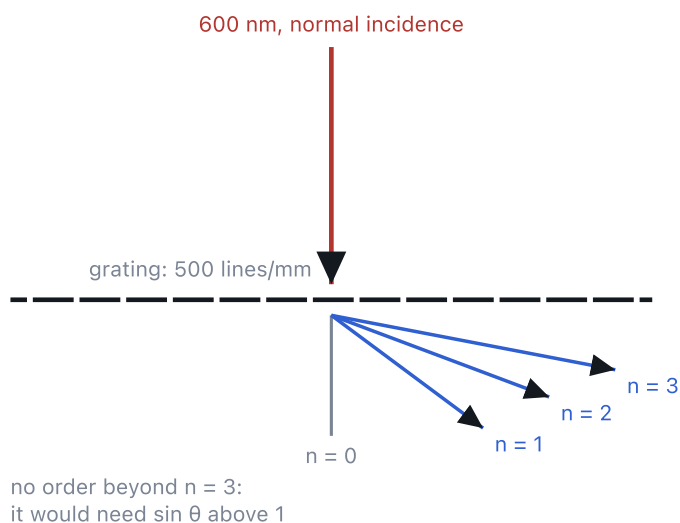


What is the strain energy stored per unit volume at the end of the straight-line region?

- A** $1.1 \times 10^5 \text{ J m}^{-3}$ **B** $2.3 \times 10^5 \text{ J m}^{-3}$
C $4.5 \times 10^5 \text{ J m}^{-3}$ **D** $2.0 \times 10^{11} \text{ J m}^{-3}$
E $4.5 \times 10^2 \text{ J m}^{-3}$

Q20 · The highest diffracted order a grating can produce **F**

Monochromatic light of wavelength 600 nm falls normally on a diffraction grating ruled with 500 lines per millimetre.

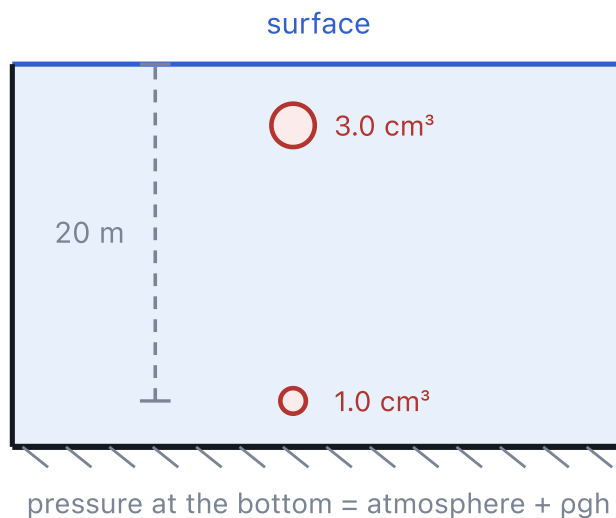


What is the highest order of diffracted beam that can be observed?

- A** 3 **B** 4
C 6 **D** 7
E 12

Q24 · A bubble rising in a lake: pressure at depth against volume **J**

A bubble of volume 1.0 cm^3 is released from the bottom of a lake 20 m deep. The atmospheric pressure at the surface is $1.0 \times 10^5 \text{ Pa}$, the density of water is 1000 kg m^{-3} , and $g = 10 \text{ N kg}^{-1}$. The temperature is the same at the bottom and at the surface.



What is the volume of the bubble when it reaches the surface?

- | | |
|-----------------------------|-----------------------------|
| A 3.0 cm^3 | B 2.0 cm^3 |
| C 4.0 cm^3 | D 1.5 cm^3 |
| E 1.0 cm^3 | |

Q25 · What survives one and a half half-lives **K**

A radioactive source has half-life **T**.

What fraction of the original nuclei remain undecayed after a time **$1.5T$** ?

- | | |
|----------------|--------------------------|
| A $1/2$ | B $1/4$ |
| C $3/4$ | D $1/(2\sqrt{2})$ |
| E $1/8$ | |